

Name: Bushya Alkarmi

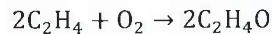
Any attempt to use an electronic device with internet access (cellphone, tablet, smartwatch, etc.) during this exam will be regarded as cheating. UIC considers cheating a serious violation of academic integrity, with penalties typically ranging from a failing grade on the exam or in the course to suspension or expulsion.

CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, CLOSED BOOK, TIME: 120 MIN

1. Ethylene oxide C_2H_4O is produced isothermally at $300^\circ C$ by the vapor-phase catalytic oxidation of ethylene C_2H_4 with air.

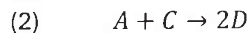
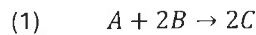


$$r'_{C_2H_4} = -k P_{C_2H_4}^{1/3} P_{O_2}^{2/3} \quad k = 4 \times 10^{-8} \frac{\text{mol}}{\text{Pa.kg.s}} \text{ at } 300^\circ C$$

The process takes place in two sequential fluidized bed reactors. The feed contains ethylene, oxygen, and nitrogen (inert) with molar flow rates of 2, 2, and $6 \frac{\text{mol}}{\text{s}}$. The pressure of the feed, the first reactor, and the second reactor are maintained at 10, 6, and 4 bar respectively ($1 \text{ bar} = 10^5 \text{ Pa}$).

Estimate the catalyst weight in each reactor, given that the ethylene conversion is 50% after the first reactor and 80% after the second.

2. The elementary liquid-phase reactions



are carried out in a 50 dm^3 well-mixed CSTR equipped with a heating jacket. A mixture of A and B is fed to the reactor at the volumetric flow rate of $1 \frac{\text{dm}^3}{\text{s}}$, temperature of 500 K and concentrations $C_{A0} = 1.0 \frac{\text{mol}}{\text{dm}^3}$ and $C_{B0} = 2.0 \frac{\text{mol}}{\text{dm}^3}$. At steady state, the outlet temperature is measured to be 450 K and the outlet concentrations are $C_A = 0.25 \frac{\text{mol}}{\text{dm}^3}$, $C_B = 1.0 \frac{\text{mol}}{\text{dm}^3}$, $C_C = 0.75 \frac{\text{mol}}{\text{dm}^3}$, and $C_D = 0.5 \frac{\text{mol}}{\text{dm}^3}$.

Additional information:

$$\text{Heat capacities: } C_{PA} = 20 \frac{\text{J}}{\text{mol.K}}, C_{PB} = 20 \frac{\text{J}}{\text{mol.K}}, C_{PC} = 30 \frac{\text{J}}{\text{mol.K}}, C_{PD} = 25 \frac{\text{J}}{\text{mol.K}}$$

$$\text{Heats of reactions: } \Delta_{r1}H = 20 \frac{\text{kJ}}{\text{mol A}} \text{ and } \Delta_{r2}H = -10 \frac{\text{kJ}}{\text{mol A}}$$

Heating fluid enters the jacket at a temperature $T_{a0} = 600 \text{ K}$, mass flow rate: $\dot{m}_a = 0.02 \frac{\text{kg}}{\text{s}}$ with heat capacity $C_{Pa} = 4500 \frac{\text{J}}{\text{K.kg}}$

- 2.1. Write the mole balance equations for all four species and estimate the reaction rates $r_{A(1)}$ and $r_{A(2)}$ within the reactor.
- 2.2. Write the energy balance equation for the reactor and estimate the rate of heat transfer into the reactor $\dot{Q}$, assuming negligible shaft work.
- 2.3. Write down the energy balance for the heating fluid and calculate the exit temperature of the heating fluid. Estimate the overall heat transfer coefficient, given the heat transfer area is $A = 120 \text{ dm}^2$.

Name: Bushra Alkarmi

Any attempt to use an electronic device with internet access (cellphone, tablet, smartwatch, etc.) during this exam will be regarded as cheating. UIC considers cheating a serious violation of academic integrity, with penalties typically ranging from a failing grade on the exam or in the course to suspension or expulsion.

$$1) \quad K = 4 \times 10^{-8} \frac{\text{mol}}{\text{Pa kg s}}, \quad r_{C_2H_4}' = -K P_{C_2H_4}^{1/3} P_{O_2}^{2/3}, \quad 2C_2H_4 + O_2 \rightarrow 2C_2H_4O$$

$$P_1 = 6 \text{ bar} = 6 \times 10^5 \text{ Pa}, \quad P_2 = 4 \text{ bar} = 4 \times 10^5 \text{ Pa}, \quad P_{\text{Feed}} = 10 \text{ bar} = 10 \times 10^5 \text{ Pa}$$

$$F_{C_2H_4} = 2 \text{ mol/s}, \quad F_{O_2} = 2, \quad F_{N_2} = 6 \text{ mol/s},$$

$$X_1 = 50\% = 0.5, \quad X_2 = 80\% = 0.8$$

$$A = C_2H_4, \quad F_{A0} = 2 \text{ mol/s}$$

$$F_A = F_{A0}(1-x) = 2(1-x)$$

$$F_{O_2} = 2 - x$$

$$F_{N_2} = 6$$

$$F_P = 2x$$

$$F_T = 2(1-x) + (2-x) + 6 + 2x$$

$$F_T = 10 - x$$

$$P_A = y_A P = \frac{F_A}{F_T} P = \frac{2(1-x)}{10-x} P$$

$$P_{O_2} = \frac{2-x}{10-x} P$$

$$\begin{aligned} -r_A' &= K \left[\frac{2(1-x)}{10-x} P \right]^{1/3} \left[\frac{2-x}{10-x} P \right]^{2/3} \\ &= KP \left[\frac{2(1-x)}{10-x} \right]^{1/3} \left[\frac{2-x}{10-x} \right]^{2/3} \end{aligned}$$

$$\begin{aligned} KP_1 &= (4 \times 10^{-8})(6 \times 10^5) \\ &= 0.024 \end{aligned}$$

Name: Bushra Alkarmi

Any attempt to use an electronic device with internet access (cellphone, tablet, smartwatch, etc.) during this exam will be regarded as cheating. UIC considers cheating a serious violation of academic integrity, with penalties typically ranging from a failing grade on the exam or in the course to suspension or expulsion.

$$2) \quad C_{A0} = 1.0 \frac{\text{mol}}{\text{dm}^3}, \quad C_{B0} = 2.0 \frac{\text{mol}}{\text{dm}^3}, \quad V = 50 \text{ dm}^3, \quad v_0 = 1 \text{ dm}^3/\text{s}$$

$$C_A = 0.25 \frac{\text{mol}}{\text{dm}^3}, \quad C_B = 1.0 \frac{\text{mol}}{\text{dm}^3}, \quad C_C = 0.75 \frac{\text{mol}}{\text{dm}^3}, \quad C_D = 0.5 \frac{\text{mol}}{\text{dm}^3}$$

$$C_{PA} = 20, \quad C_{PB} = 20, \quad C_{PC} = 30, \quad C_{PD} = 25, \quad C_{PA} = 4500 \frac{\text{J}}{\text{K} \cdot \text{kg}}$$

$$\Delta r_1 H = 20, \quad \Delta r_2 H = -10 \frac{\text{KJ}}{\text{mol}_A}, \quad T_{A0} = 600 \text{ K}, \quad \dot{m}_A = 0.02 \text{ kg/s}$$

2.1) at steady state:

$$\text{in} - \text{out} + \text{generation} = 0$$

$$r_1 = r_{A(1)}, \quad r_2 = r_{A(2)}$$

$$v_0 C_{A0} - v_0 C_A = V(r_1 + r_2)$$

$$(1)(1) - (1)(0.25) = 50(r_1 + r_2)$$

$$r_1 + r_2 = 0.015$$

$$v_0 C_{B0} - v_0 C_B = V(2r_1)$$

$$(1)(2) - (1)(1) = 50(2r_1)$$

$$r_1 = 0.01 \frac{\text{mol}}{\text{dm}^3 \cdot \text{s}}$$

$$0.01 + r_2 = 0.015$$

$$r_2 = 0.005 \frac{\text{mol}}{\text{dm}^3 \cdot \text{s}}$$

Name: Bushra Alkarmi

Any attempt to use an electronic device with internet access (cellphone, tablet, smartwatch, etc.) during this exam will be regarded as cheating. UIC considers cheating a serious violation of academic integrity, with penalties typically ranging from a failing grade on the exam or in the course to suspension or expulsion.

$$2.3) \quad \dot{Q} = \dot{m} c_p (T_{Ao} - T_{Bo})$$

$$4500 = (0.02)(4500)(600 - T_{Bo})$$

$$T_{Bo} = 550 \text{ K}$$

$$A = 120 \text{ dm}^2 = 1.20 \text{ m}^2$$

$$T_R = 450 \text{ K}$$

$$\Delta T_1 = 600 - 450 = 150 \text{ K}$$

$$\Delta T_2 = 550 - 450 = 100 \text{ K}$$

$$\Delta T_{in} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}$$

$$= \frac{150 - 100}{\ln(150/100)}$$

$$= 123.3 \text{ K}$$

$$\dot{Q} = UA \Delta T_{in} \rightarrow U = \frac{\dot{Q}}{A \Delta T_{in}}$$

$$= \frac{4500}{(1.20)(123.3)}$$

$$= 30.4 \frac{\text{W}}{\text{m}^2 \text{K}}$$